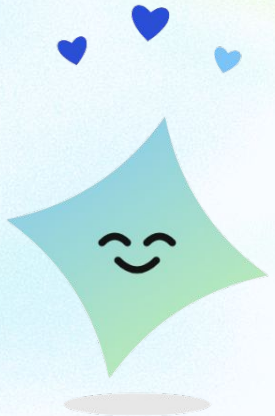


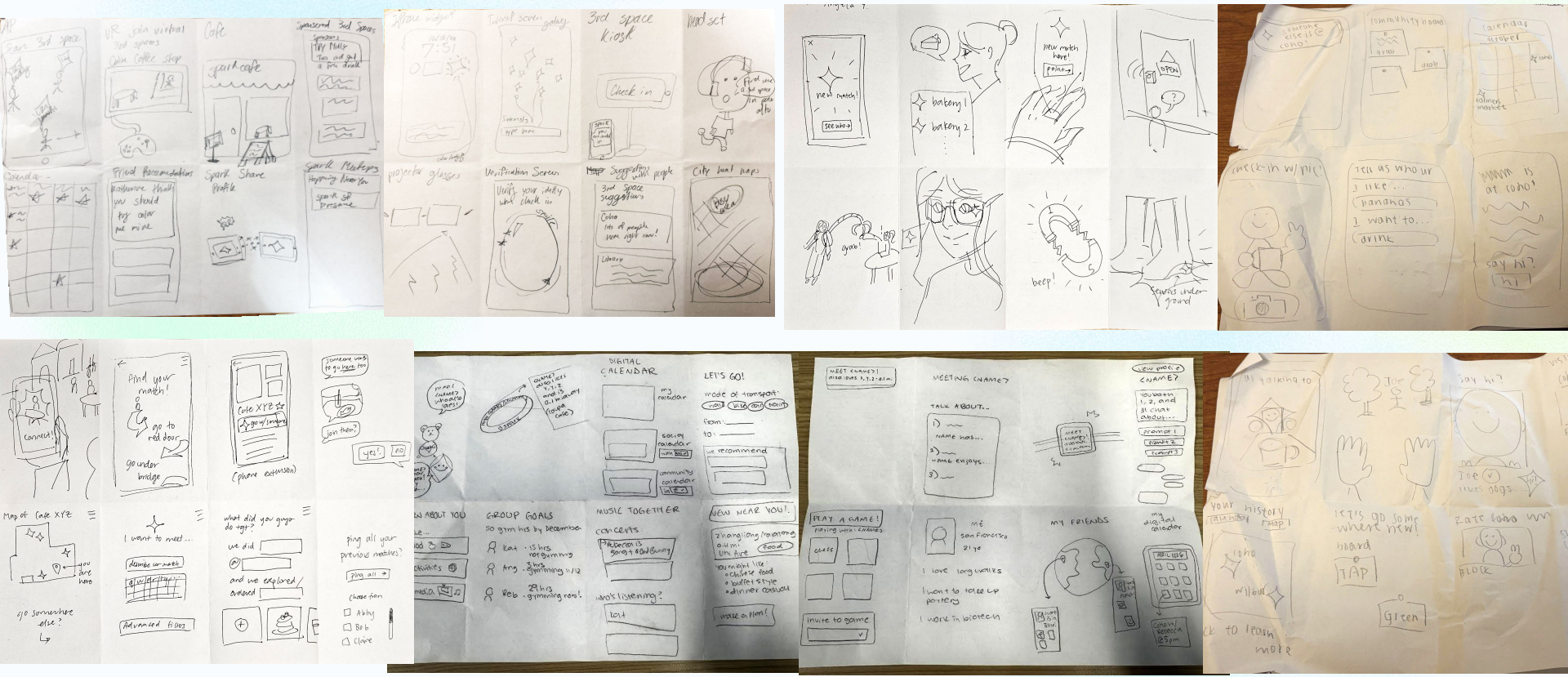
by the KRAMberries

(Katherine Kricorian, Rebecca Alcala, Angela Yang, Meredith Nam)

sp rk-ing connections at third spaces with one click.

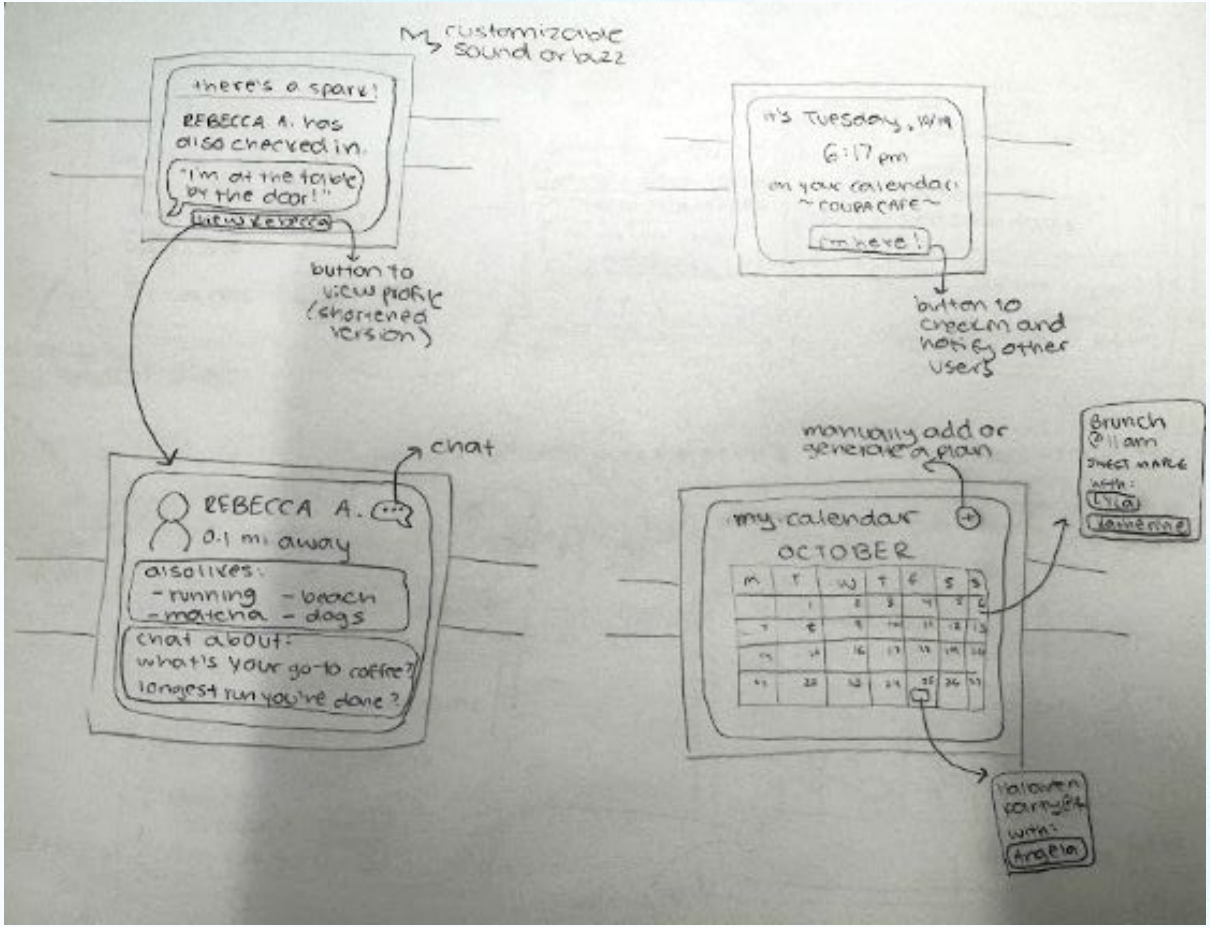


brainstorming solutions



Crazy 8s

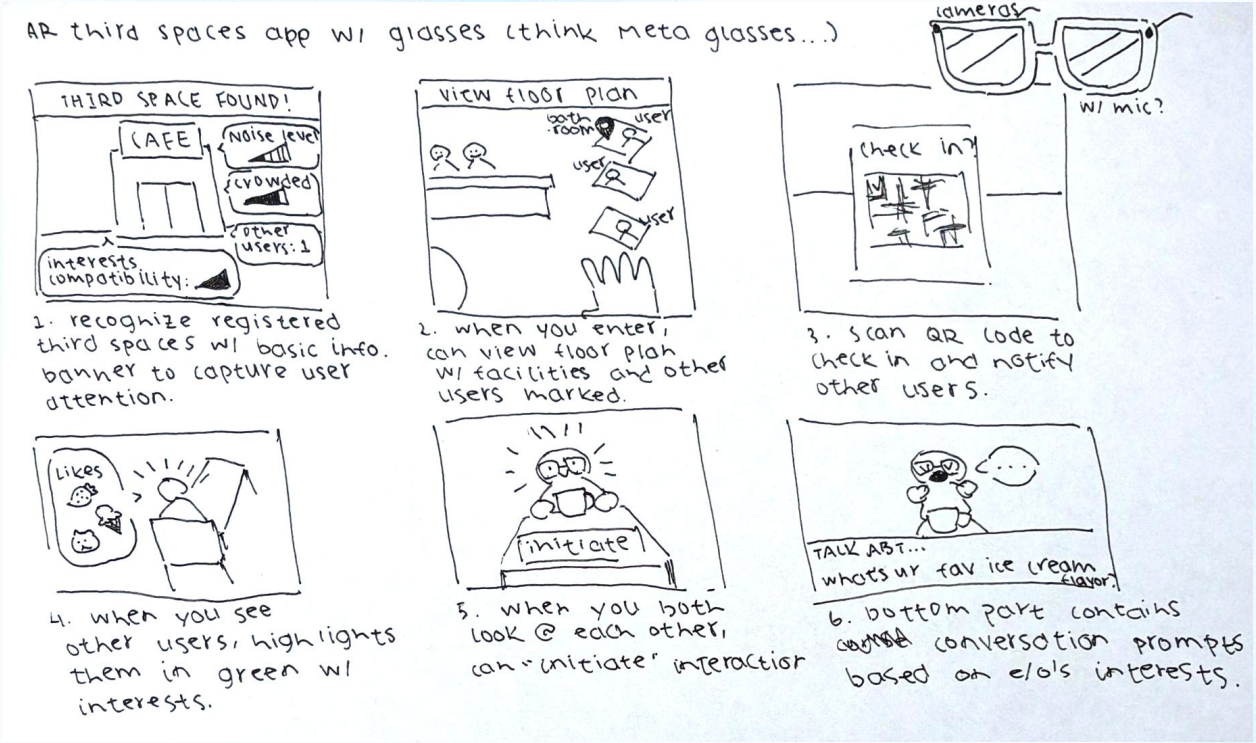
our apple watch concept sketches



APPLE WATCH

A wearable watch that notifies the user when a **potential friend is nearby** and enables them to **easily check in** at third spaces.

our AR concept sketches

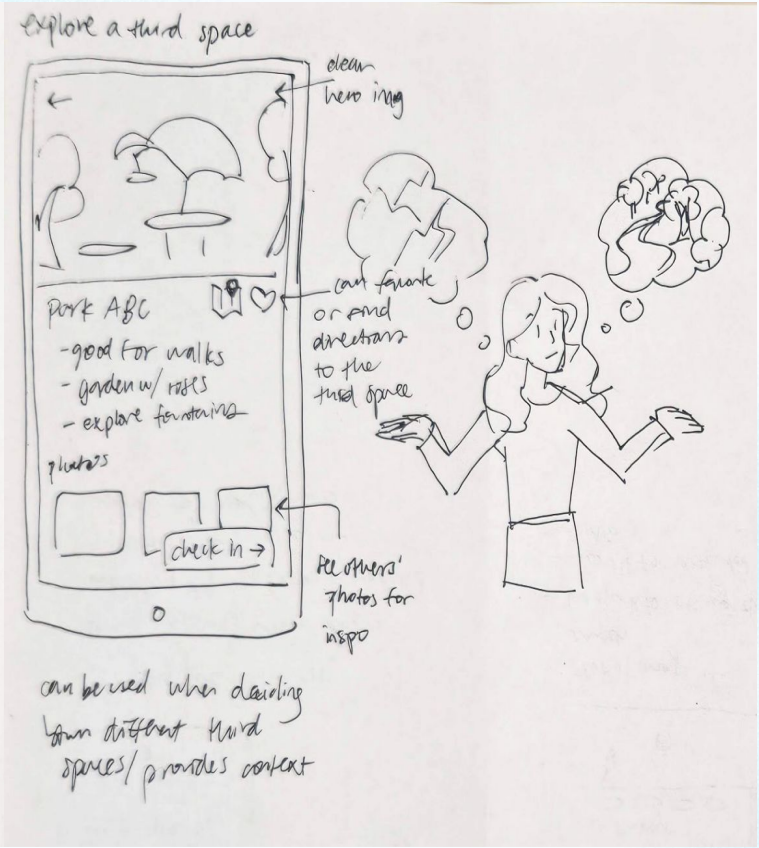


AR APP built into smart glasses. **spotlights third spaces**, with relevant info and predicted suitability to user, **and other app users**, with info about interests and conversation starters. **users check in to third spaces w/ glasses**.

our mobile app concept sketches



Setting up user profile

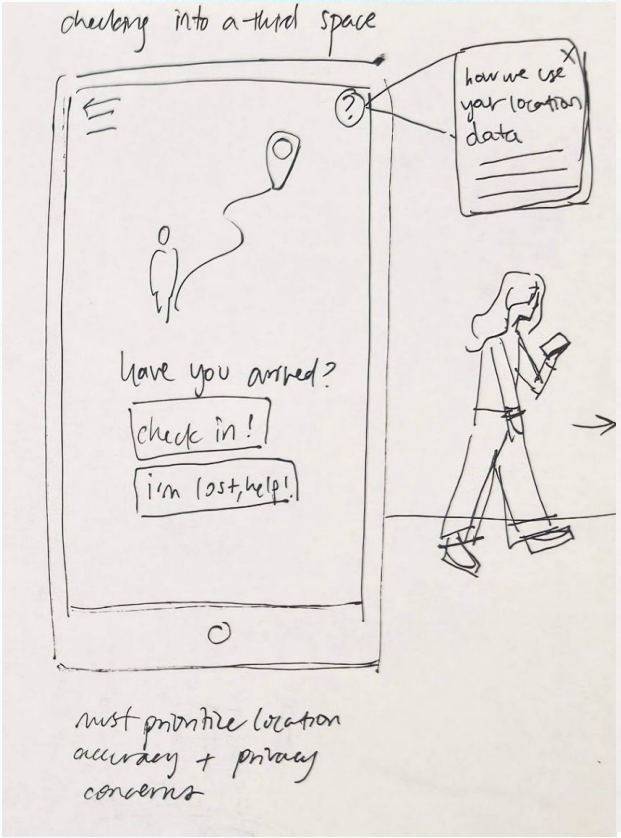


Exploring third spaces

MOBILE APP

A phone app that lets users take **personality quizzes/assessments,** choose to check in at third spaces, and possibly connect to **location-based apps** (rideshare, maps)

our mobile app concept sketches



Checking into a third space

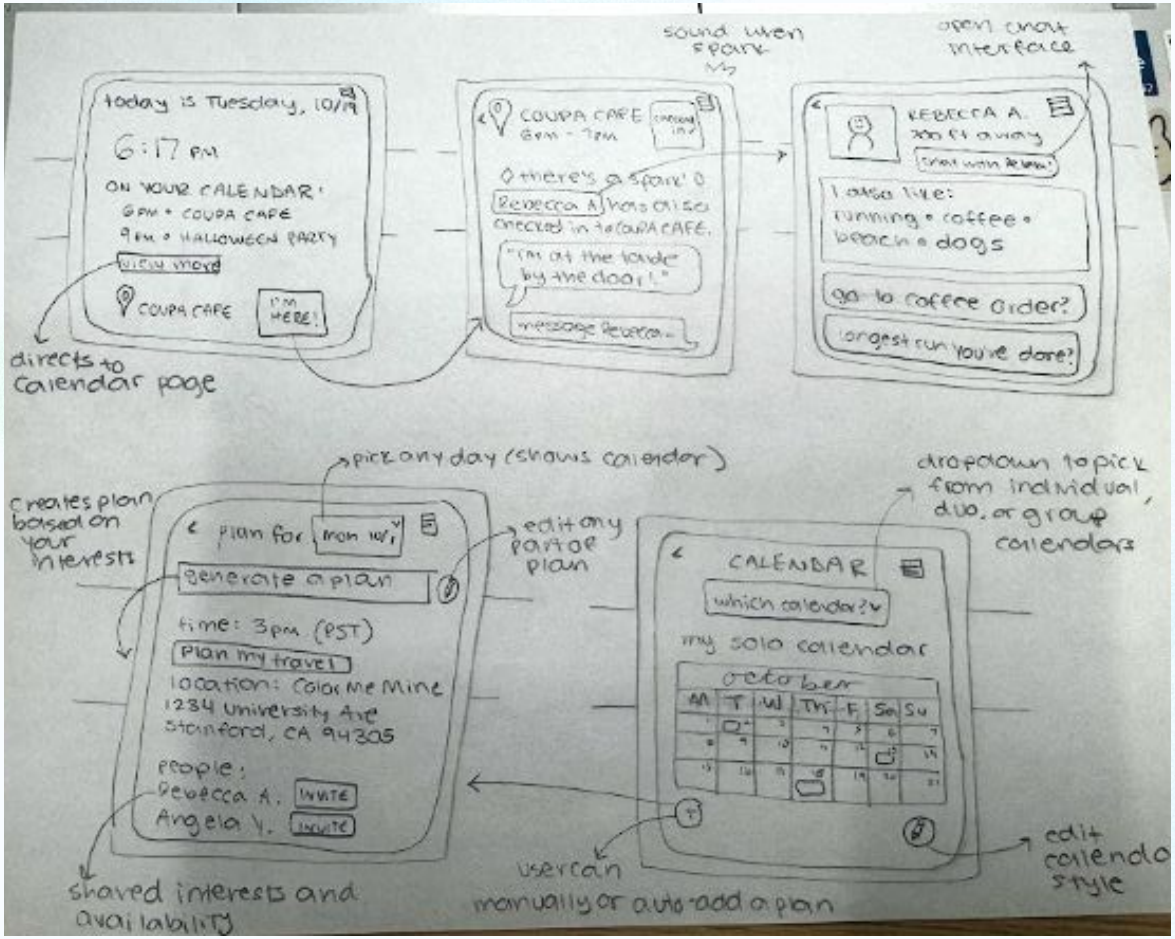


Getting a match

MOBILE APP

A phone app that lets users take **personality quizzes/assessments,** choose to check in at third spaces, and possibly connect to **location-based apps** (rideshare, maps)

building on our apple watch realizations

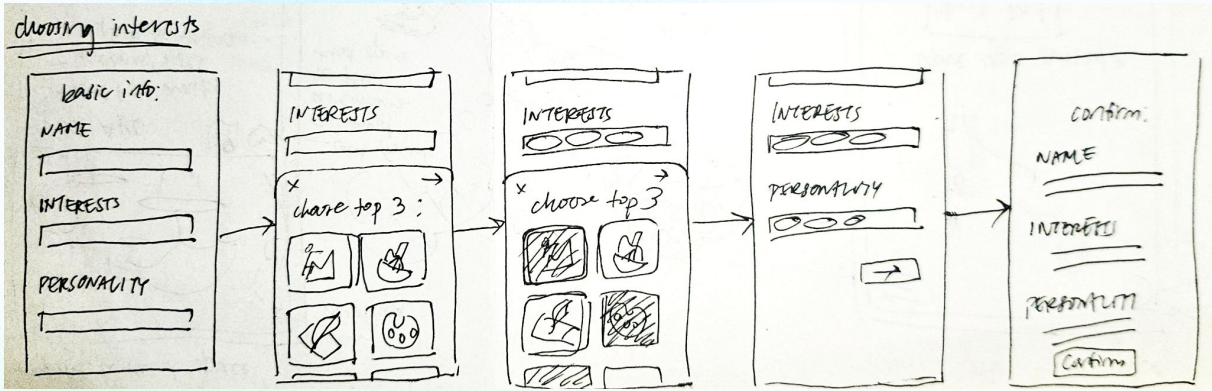


View plans on your calendar

Check in easily when at a third space

Get notified when a potential friend is nearby + reference talking points

building on our mobile phone app realizations



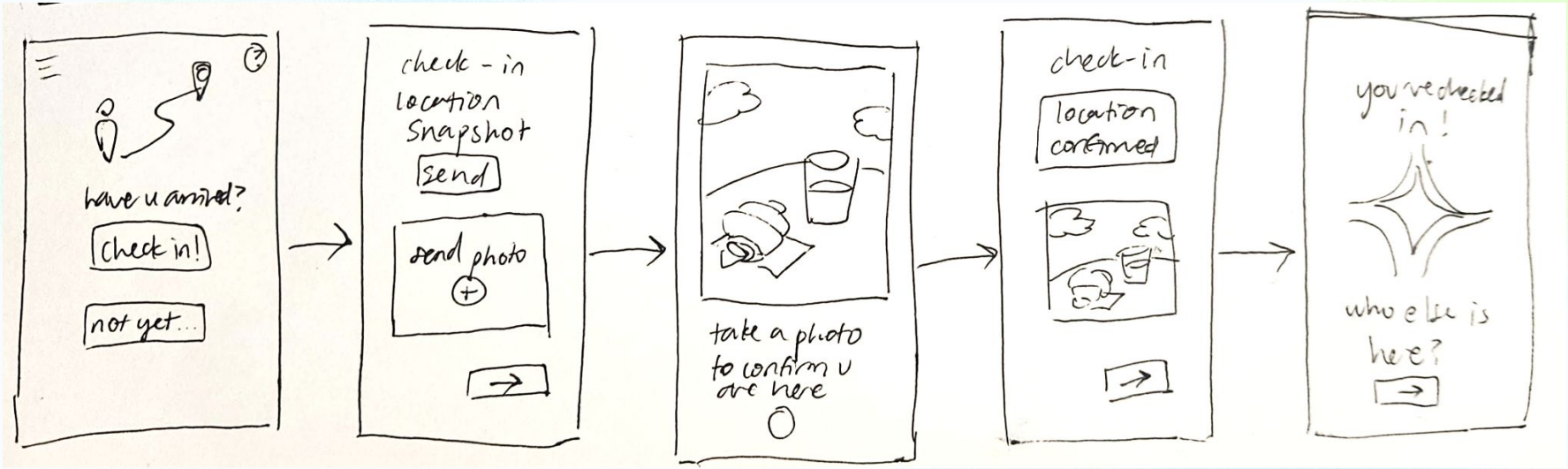
Setting up user profile



Exploring other third spaces

building on our mobile phone app realizations

Checking into a third space



Choosing our final realization: mobile app

APPLE WATCH PROS

- worn on wrist, tactile vibrations effective to **“nudging” users to act** on real-time notifications
- **less distracting/imposing** than phone screen for real-world conversations

APPLE WATCH CONS

- small screen -> **hard to input/view detailed information** but app relies on detailed user info. potential **accessibility issue** w/ visually-impaired
- wearability useful when user in third space, but **may not reach this stage** if users quit b/c inputting/viewing info is **too hard early on**

MOBILE APP PROS

- very accessible (6-7x more iPhones sold than Apple Watches), **lower learning curve**
- **familiar format** for our target users (young professionals and users in social hubs/cities
- apple watch is only within the apple ecosystem, while mobile app can be **accessed on any brand of phone**

MOBILE APP CONS

- users may conflate our app w/ **social/romantic “matching” apps** like Tinder or Bumble on app store
- phones are **bulkier and inconvenient** to “whip out” during a conversation, and may **make users appear disengaged**

Choosing our final realization: mobile app

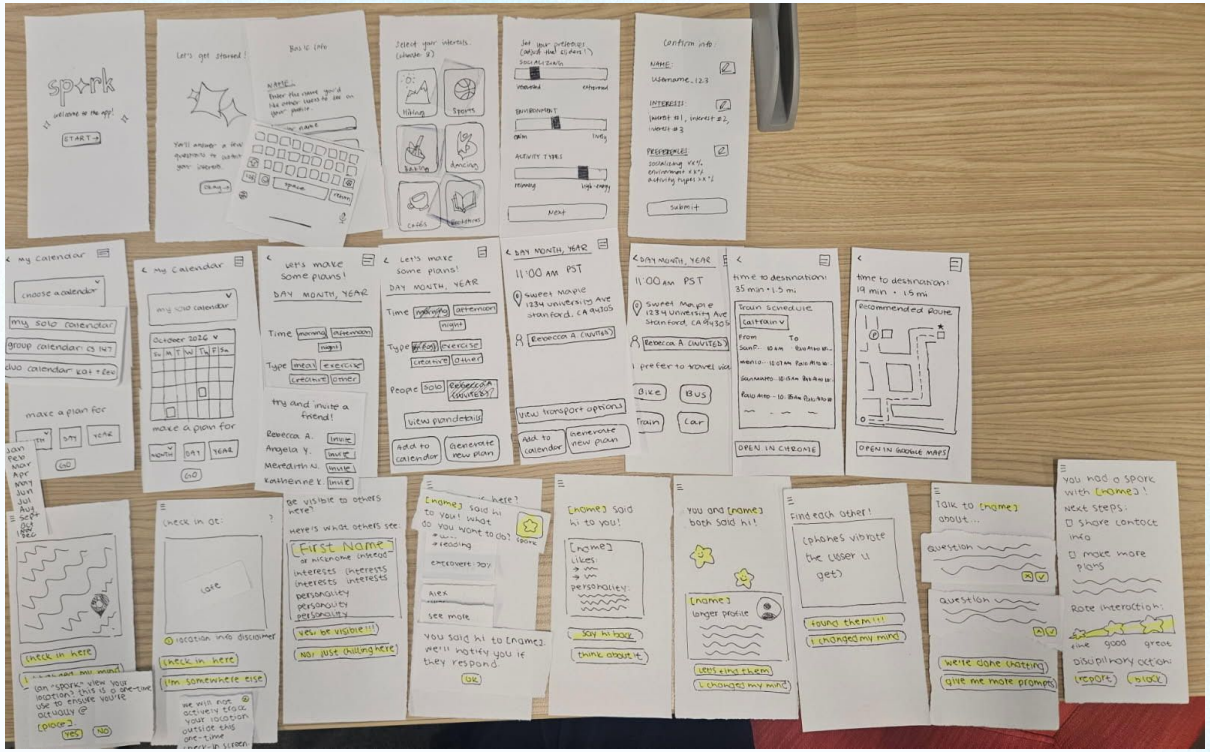
WHY IS A MOBILE APP SUPERIOR?

- regardless of time/place, most adults carry their phone, so app is **readily accessible**
- app relies on **user input/personalization**, accessibility features built into most phones
- sophisticated features (location check-in, maps/rideshare integration) are common in existing apps
    - **sufficient technical support** and user familiarity in phones
- actions required in our app such as **taking photos, scanning ID f**or verification, etc. most **compatible w/ phone cameras**

CONSTRAINTS OF OUR PLATFORM

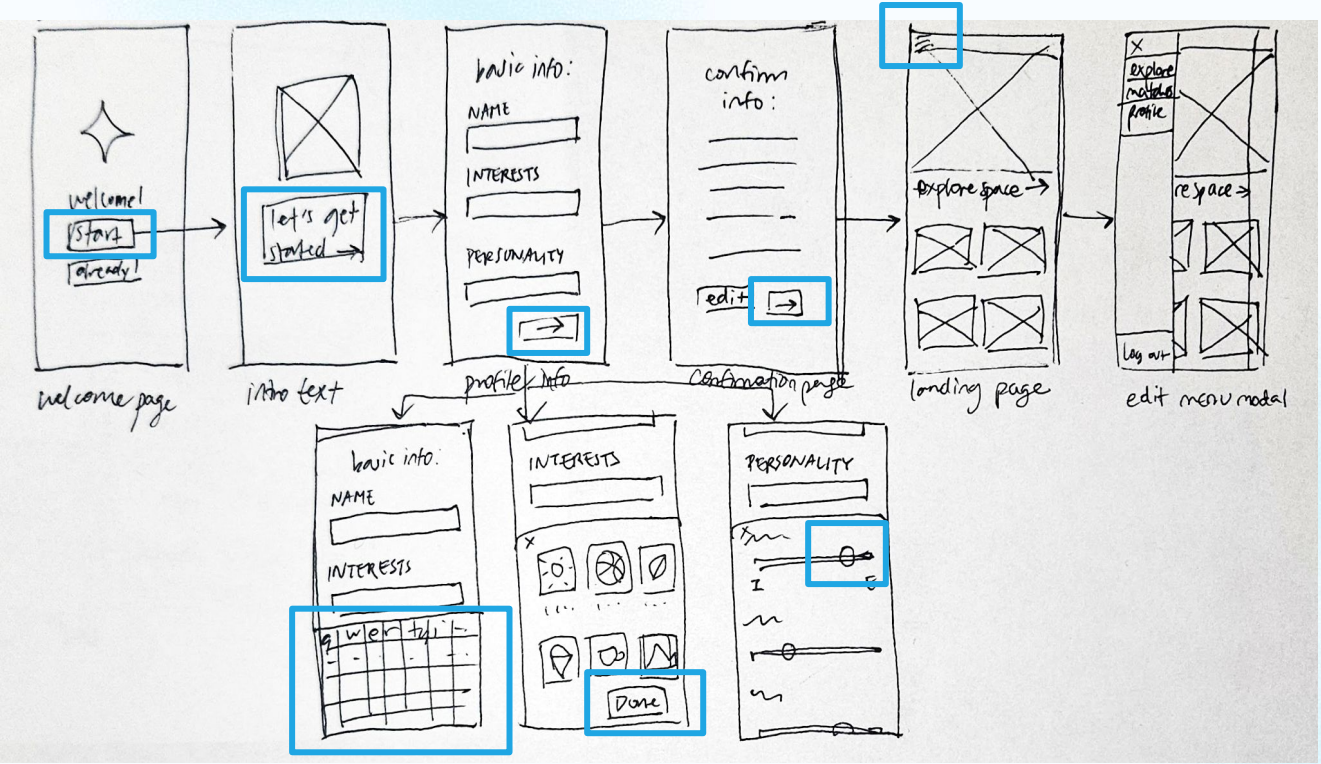
- easy to **dismiss notifications** from phone or forget newly downloaded app
    - difficult to initiate third-space meetups if users **disable vibrations/notifications** and close app
- inconvenient to quickly access app w/out active thought, esp. when we're trying to **seamlessly integrate third space interactions** into daily routine
    - user must react to notifications OR remember app, take out phone, and open app before using

Making our lo-fi paper prototype



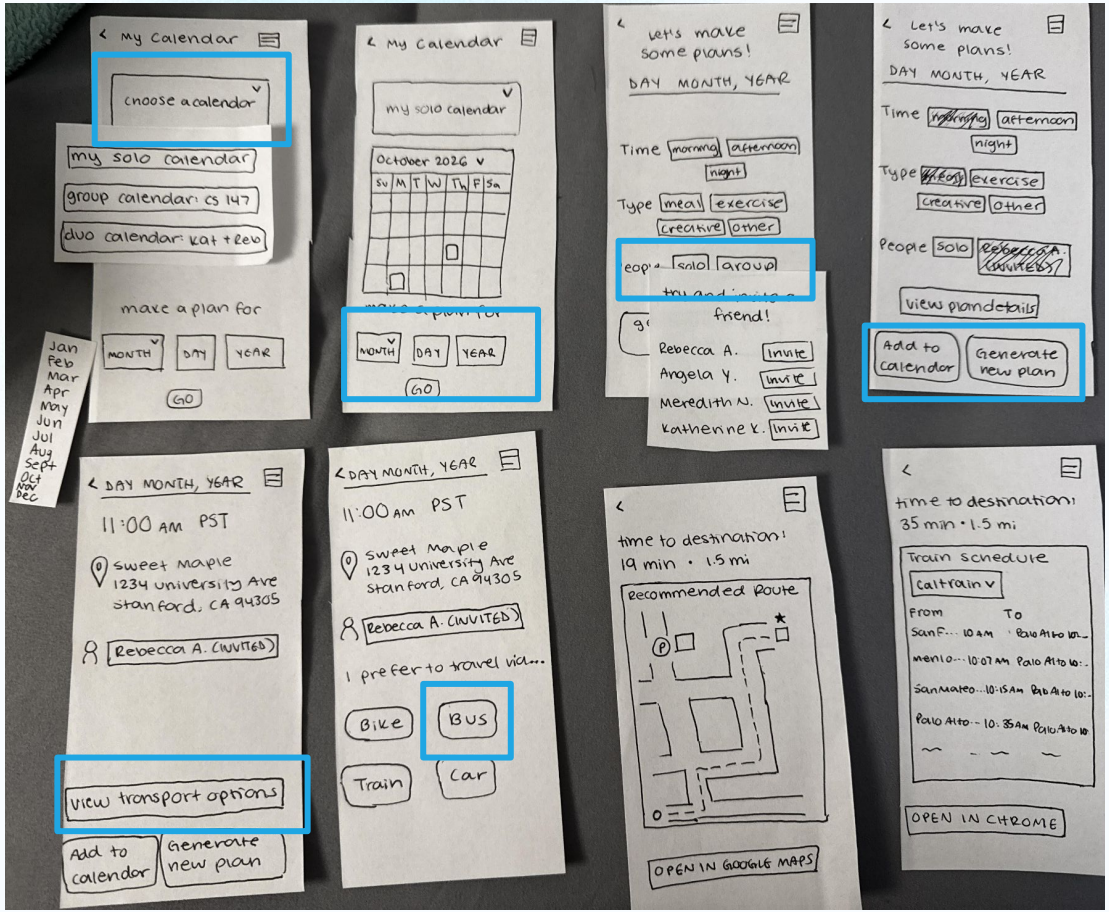
- built in 3 components **separated by the 3 tasks** to make sure all were clear
- main features: **buttons** that triggered next page, **fill-in-the-blank, sliders**

Lo-fi prototype - simple task



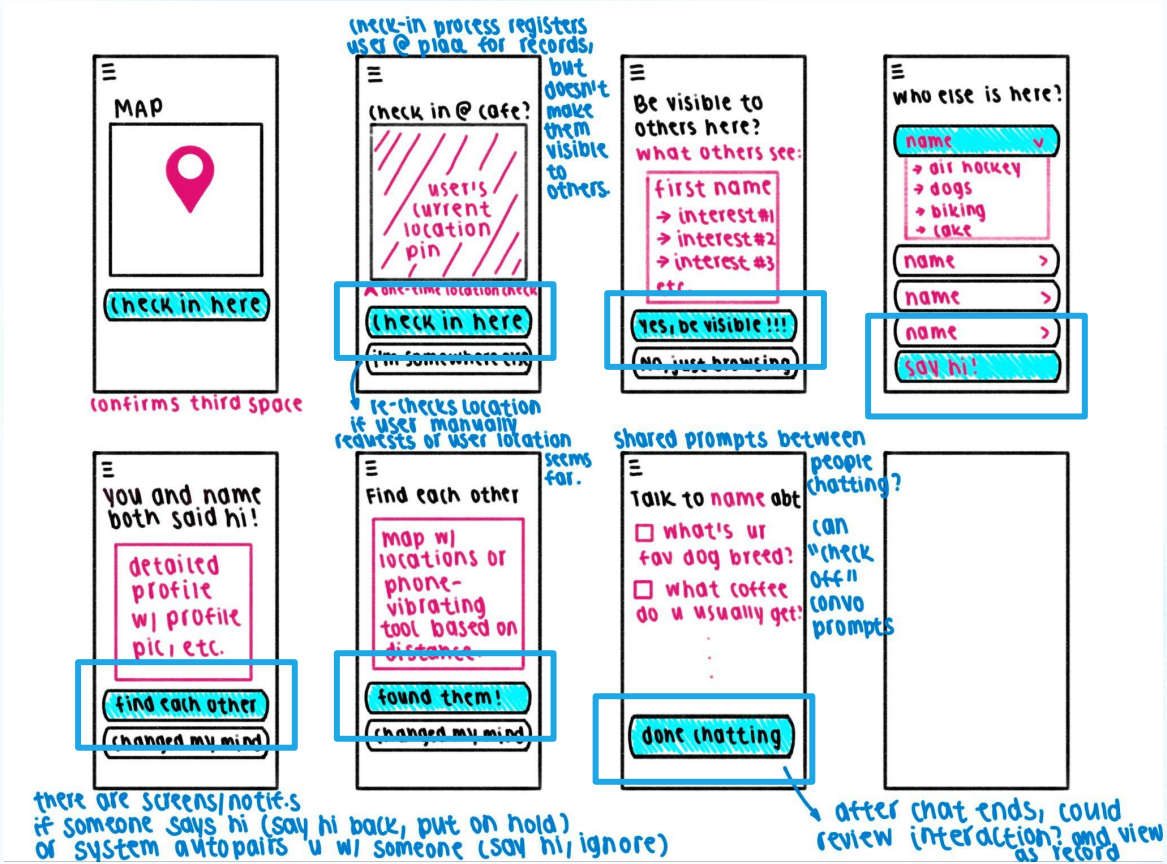
1. *User logs in for the first time*
2. *Enters basic info & hobbies*
3. *Navigates to landing page*

lo-fi prototype – moderate task



1. User selects a calendar
2. Enters a date to make a new plan
3. Indicate plan preferences and invite prior friends
4. View generated plans
5. View transportation options

lo-fi prototype – complex task



1. *User arrives at the third space*
2. *Check in*
3. *Choose visibility preferences*
4. *View others who have checked in*
5. *Direct to potential "sparks"*
6. *Rate the interaction*

testing methodology — participants

Alex Stanford student (20)

- very **extroverted** and eager to engage with us, encountered alone
- we reached out to them **passing by in the hallway**
- **CoDa** after class, setup in a room

Beth & Bill Working adults touring Stanford (30s)

- quieter and **hesitated to test prototype**, but became less guarded as testing continued
- found walking outside of CoDa
- setup at **outdoor tables (in the rain)**



Carlos Visiting from San Jose (19)

- found doing work @ CoHo
- quiet, seemed engrossed in his work, so **harder-to-approach** impression
- fits target audience of **more introverted, less adventurous i**ndividuals less likely to initiate interactions themselves
- setup at CoHo table

Daniel Visiting from San Jose (20)

- found doing work @ CoHo
- **enthusiastic**, asked more questions, seemed **engaged and curious** about the future of the app
- setup at CoHo table
- interviewed after Carlos

testing methodology — environment & procedure

Katherine: note-taker (wrote down notes based on auditory environment)

Rebecca: computer (moved pieces according to user actions)

Meredith: observer (took photos and observed physical interactions w/ and reactions to prototype)

Angela: facilitator (gave initial background information and task descriptions)

OUR PROCEDURE

1. identified and approached potential tester
2. described project and app concept
3. gave first (simple) task to complete
4. answered questions/direction with other tasks



testing methodology — usability goals & measurements

Efficient

measured by **# of times each task takes <5 min**

- users can quickly create a profile, make plans, and check in to third spaces to increase focus on in-person actions like **interacting** w/ others

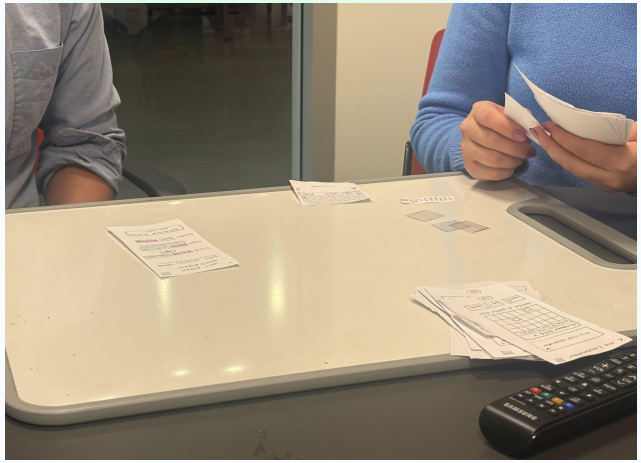
Pleasing

measured by **numerical rating for mock "interaction"**

- want people to explore as many third spaces and meet as many ppl as possible b/c app relies on active users and updated info abt specific third spaces

testing results – Alex





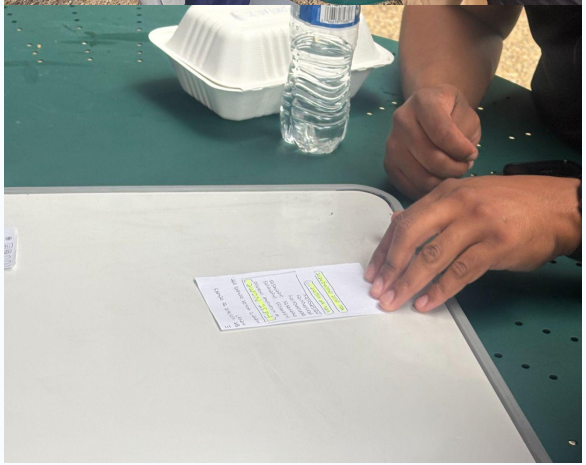
- intuitive understanding of personality page, enjoyed **customizing and adding preferences**
- confused at the **layout of the Calendar Selection** page
- unclear if the “suggested friends to invite” were **new or old**
- confused on **wording of certain buttons**, had to clarify meaning

Usability Goals

- ***efficient***: indicated plan preferences quickly and **almost immediately** met up with a friend (~5 min plan generation, ~5 min check-in process)
- ***pleasing***: enthusiastically gave highest rating for interaction, satisfied with customizing user profile but w**anted visible proof** that user profile influences third space recommendations

testing results – Beth & Bill



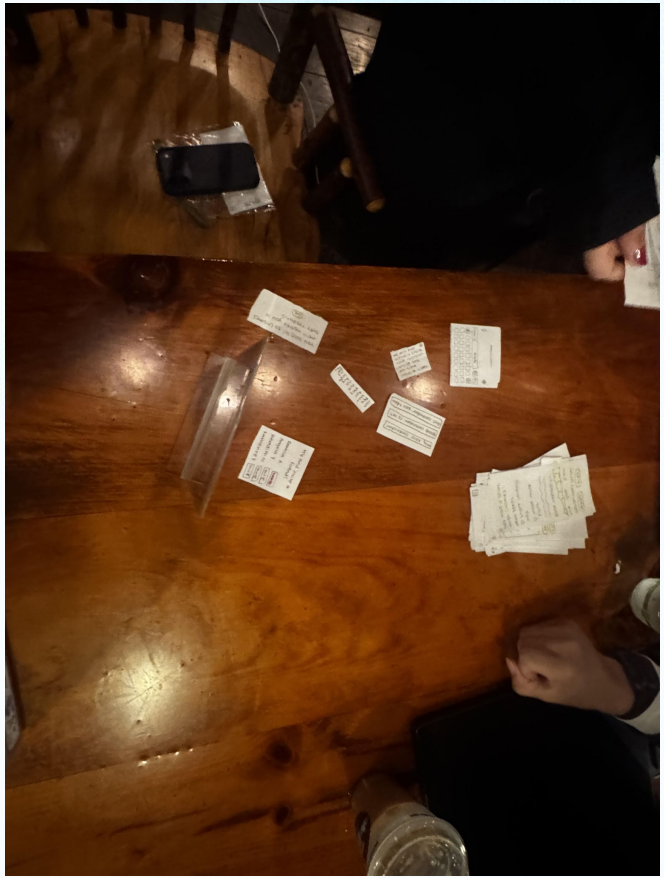


- selected the majority of the **default options**
- very curious and **clicked all the buttons** on the screens (including disclaimers and options)
- curious about l**ocation and privacy preferences**

Usability Goals

- ***efficient***: went through some parts **quickly (~5 min check-in process)** but spent more time reviewing location disclaimer – balance of efficient and option to slow down
- ***pleasing***: enjoyed **customizing personality** and reflecting on which options best suited them, excited at the **prospect of meeting new people** (“this can be your new Tinder”), enthusiastically selected highest star rating

testing results – Carlos

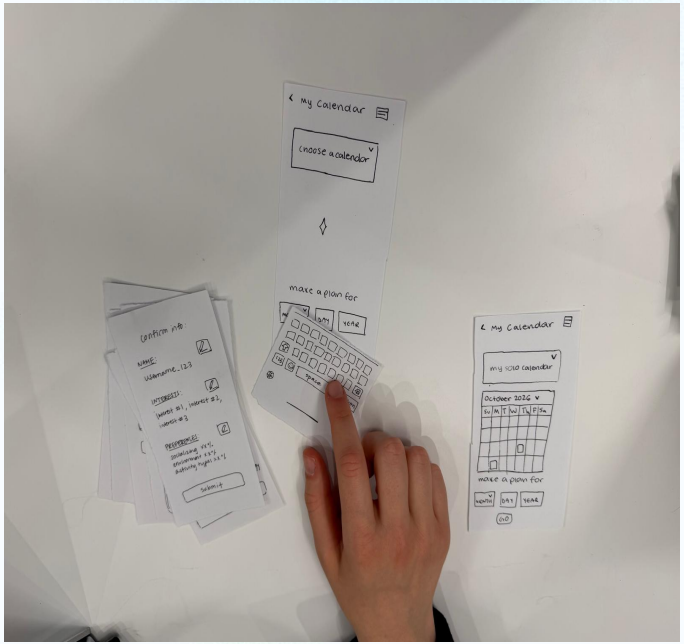


- curious to **where the friends came from** on the screen (friends you have on the app or people you have previously met up with)
- didn't follow the Calendar Selection screen/**didn't catch that it was a dropdown** to different calendars

Usability Goals

- ***efficient***: familiarity made it easy to use, concept had aspects that he **connected to other apps** ("It's like Find My Friend"), took most time figuring out the location check-in feature and **didn't seem to grasp the face/ID** scan screen
- ***pleasing***: "I would use this", enthusiastically selected highest star rating

testing results – Daniel



- liked how much he was able to **personalize his profile**
- curious to see how the app would work even **without having multiple people checked-in** to a location
- disliked **the amount of time** expected to get to the destination

Usability Goals

- ***efficient***: understood app concept quickly, spent the most time **viewing others who said "hi"** before choosing who to meet up with (~7 min)
- ***pleasing***: seemed excited to pick people to meet, **enthusiastically gave highest rating** for interaction

implications

Alex - student

- need for influence over 3rd space suggestion, users wanted to **filter spaces based on proximity** and price points
- for simplicity, we “forced” participants to go to Sweet Maple and be able to talk to 2 people there, but they seemed a bit **disgruntled w/ these options**



Beth & Bill - working adults

- preferred **simplicity and efficiency:** “just pick whatever option lets us see the next thing”
- although genuinely curious about exploring third spaces and meeting others, they have **busy schedules**
    - Want working adults to feel like their spare time is **used meaningfully**, not wasted
- read location confirmation & disclaimer out loud, so **they were aware when/what abt their location info** was recorded and were comfortable w/ sharing that info



implications

Carlos - student/young adult

- general structure of app is intuitive because it **mimics some parts of other apps**/concepts our main user base is familiar with (Find My Friend)
- calendar selection was confusing, so **browser-like tabs** instead of buttons clearly differentiate and highlight them



Daniel - student/young adult

- people crave ***more* info about others** in third spaces before they meet
    - balance **initial privacy/anonymity** w/ enough info to make users interested in meeting but not worried bad actors will abuse their personal info
- b/c our app hinges upon **convenience and efficiency,** third-space recommendations should allow users to **filter by proximity**



what this means

design changes

- **should feel effortless w/ simple, readable options**
  - personalization process should be seamless so users don't quit before they visit a third space
- **revise language to be consistent throughout (people got confused by “say hi”), play on “spark” language.**
- users should feel full control over how their interests shape event plans / they find event plans **in a meaningful way**, rather than the app providing generic recommendations

testing limitations

- **didn’t test with older individuals**, one of target audiences, unsure if they have distinct UX/UI needs
- **reliability of “efficiency” as usability goal**
  - moderate task presets all third space options, so users were forced to pick 1 option and couldn’t explore others, automatically making the task efficient

thank you!

- sincerely, the KRAMberries



appendix:

Full pros & cons

APPLE WATCH PROS

- easily take in quick, glanceable information w/out user action
- b/c it's worn on wrist, vibrations are very effective to "nudging" users to act on real-time notifications
- Wearable and easy to keep on hand throughout the day
- Accessible/already used by primary user base (young adults)
- Smaller screen prevents user from getting wrapped in the device and keeps them engaged in physical world/third space
- Users who are active/use watch for sports may be especially likely to download, could get a stronger base of users looking for exercise activity planning

APPLE WATCH CONS

- small screen makes it difficult to input and view detailed information, esp. when app relies on detailed user info. potential accessibility issue w/ visually-impaired individuals
- unique capabilities mostly useful when user is in third space, the only time they don't need to input/view detailed info, but user may not reach this stage if they quit b/c inputting/viewing info is too hard
- people may get distracted by their small watch screens, with vibrations/dings taking them out of the in-person socialization
- May be lost easily due to small size, compared to phones which are always on hand
- Users are much less likely to buy an Apple Watch than a new phone, less likely to already have a watch
- Notifications that are so visible may annoy users

Full pros & cons

MOBILE APP PROS

- more accessible to people (6-7 times more iPhones sold than Apple Watches)
- lower learning curve than watch application, as main functionality exists through phone screen, not other buttons
- Allows for easy integration of other phone features (signing up with phone number, camera, location services, etc.)
- Larger screen for people to be able to use and experience onboarding in
- Integrates well into existing phone-centered lifestyle
- More easily connect to other phone apps such as Google Maps, Lyft which may not be installed/used on an Apple Watch
- Could allow for more usage among a younger user base (eg. late teens, early twenties) who are more likely to already own a phone than a watch (while mid 20s or 30s may be more likely to have a watch)

MOBILE APP CONS

- users may conflate our app w/ other social/romantic “matching” apps like Tinder or Bumble on the app store
- limitations to interactive/engaging nature of third-space meetups, b/c users may disable device vibrations or notifications
- centering social interactions around devices conflicts with the serendipitous, in-person connections we’re trying to cultivate
- May be challenging for users unfamiliar with apps and phones
- Takes more effort to check notifications/make plans vs looking down at your wrist
- Phones usually receive many notifications, could be harder to distinguish ours from others if no immediate visual
- Phones are used more broadly so user may get distracted with other notifications or apps

Testing prep docs

Environment & Apparatus:

- Spaces around campus that draw in a large group of people (both from Stanford and non Stanford participants)
- Use computer to take notes, and observations during the test
- Phones to take pictures
- Paper prototype to move through as we go through the task experience with our participants

Determine team member roles:

- Include a thorough description of the procedure with images of the test being conducted. Team member roles and descriptions should be clearly outlined in this section. For example, facilitator, observer/note-taker, or computer
    - Katherine: Note-taker
    - Rebecca: computer
    - Meredith: observer
    - Angela: facilitator

Testing Scripts

“Hi, my name is [insert name] and my team and I are in Stanford’s CS147 class. We are working on a project and are in the planning stage of developing an app to help users spontaneously meet people with similar interests in third spaces.

Is it okay if we take notes and a picture of you/your hands (face or no face) interacting with the app?”

We have our paper prototype here: feel free to interact with the prototype by clicking buttons like you would on a phone. When you click a button, [insert name of facilitator] will adapt the prototype to the appropriate screen based on your action.”

Testing Scripts Pt. 2

“Please speak your thoughts aloud! Feel free to narrate any thoughts or things you like/don't like as you are going through the app.

Our goals are:

- Help people share their specific interests and personality type with others
- Help people find an interesting third space to explore in their free time
- Help people initiate meaningful conversations and spark connections with others in third spaces
- Our app lets users input their interests, automatically create plans with others with similar interests, and notifies users when someone they might get along with is nearby and is also interested in making a friend.

Please use this paper prototype as you would a regular app on your phone. Once you click a button or input text, [insert name of computer] will move to the appropriate screen and or make sure the right pop-up comes up.”

Testing critical incidents

1 - For one of our tests, we conducted it outside on the CoDa patio. Midway through the tests, it began to rain causing the team to have to adapt and find a space that would:

- Make sure the participants were still comfortable and safe from the rain
- Ensure our paper prototype would not get wet
- *Cosmetic problem: raindrops on the paper prototype but nothing damaged*

With this test, our flow was interrupted and realized that the participant's focus shifted. Before they were very interested in learning and playing with the application, but when the rain started we noticed their demeanor change and them wanting to get over the experience but quickly clicking through the applications (one participant made about comment to "just click yes so this can wrap up faster")

2 - For one screen, we had a limited number of "matches" offered. The user (who ranked themselves as very extroverted) actually felt less motivated to reach out to someone new because they felt like they wouldn't "click" with any of the example matches.

- *Minor usability problem: slowed down user's process and caused some frustration, but user still connected with their example match without many roadblocks*

Despite how introverted/extroverted users perceive themselves as, the kinds of people in their current third space matter just as much as the user's personality. Going forward, we should try to implement features that bring together users with broad interests into the same third spaces.

Testing critical incidents Pt. 2

3 - Our initial paper prototype rushed over the actual “meeting your match” user flow a bit, which led to confusion during the screen listing out suggested questions the user could talk about with their match.

- *Major usability problem: the user skipped over screens that were meant to encourage more social interaction by suggesting questions, since it wasn’t clear to them*

This made us realize that it’s crucial for our app to continuously nudge users to engage in off-screen social activity, instead of having the majority of their experience within a phone screen. We can do this by adding features that block out other distracting apps on the user’s phone and making it visually clearer that questions are meant to spark social interactions.

3 - For our first interview we did not preface what the app was about or the tasks we were looking for the user to complete. While this truly let us to understand how people would interact with the app and understand the intent of our UI without context, the user did not intuitively understand exactly what the set-up process was trying to understand about them or why it needed to be so in detail.

- *Major usability problem: user hesitated to click anything on the screen because they weren’t sure what multiple buttons meant, which halted the user flow*

To address this, we changed our initial approach and decided to better introduce a broad overview of the application and task that would be asked from them to be completed during the experience.